Lecture 3

E-Mail Services

Electronic Mail: SMTP, POP, and IMAP

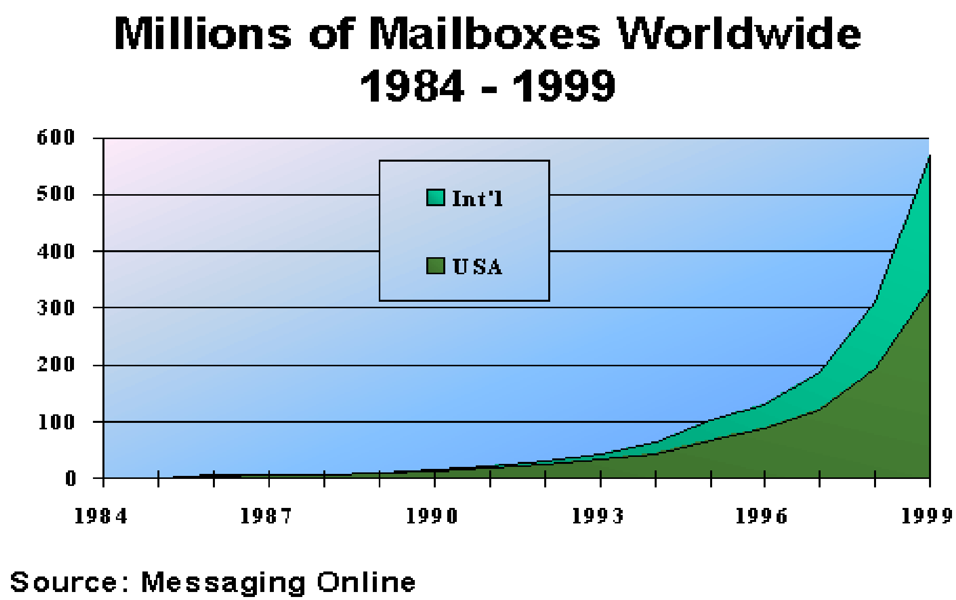
- *Understand the functions and formats of a user agent*
- *Understand MIME and its capabilities and data types*
- *Understand the functions and commands of an MTA*
- *Understand the function of POP3 and IMAP4*

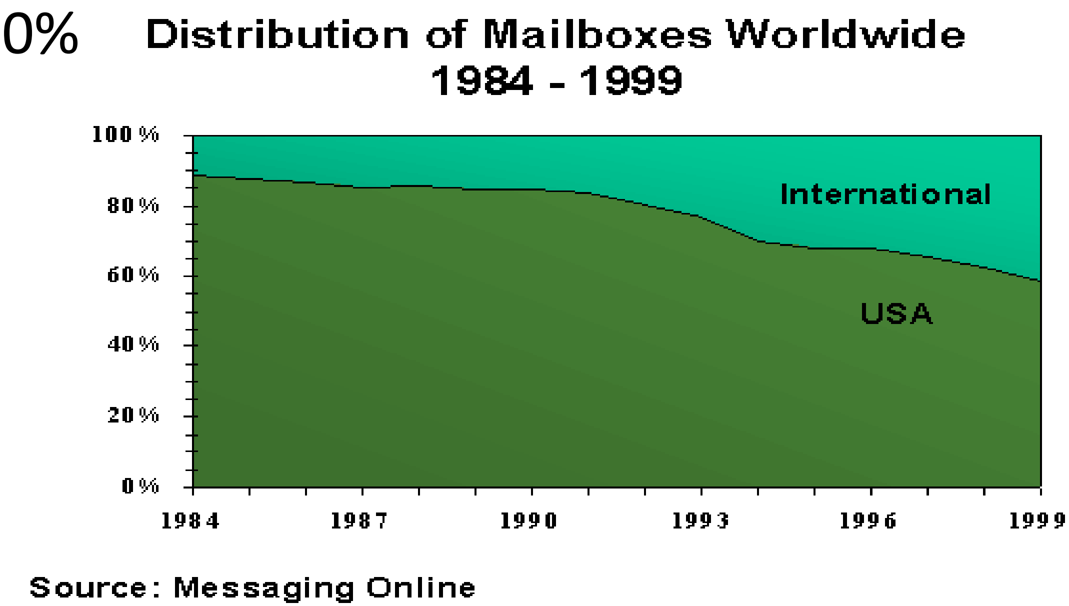
Email Statistics

Email Statistics (2004):

- 31 billion emails sent daily, expected to double by 2006
- Email generates about one billion Gigabytes of new "information" per year
- Spam accounts for about 40% of all email traffic

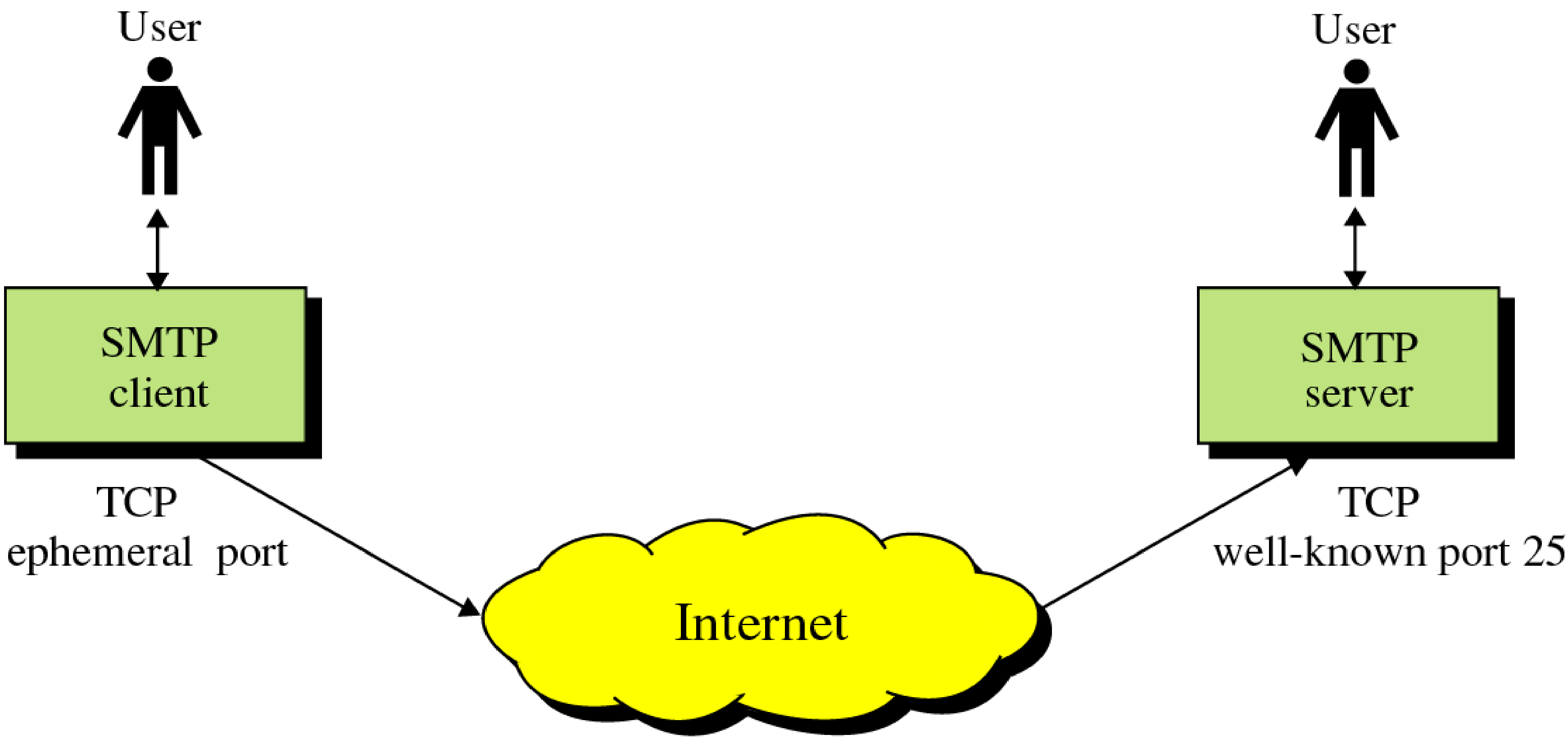
http://www.spamfilterreview.com





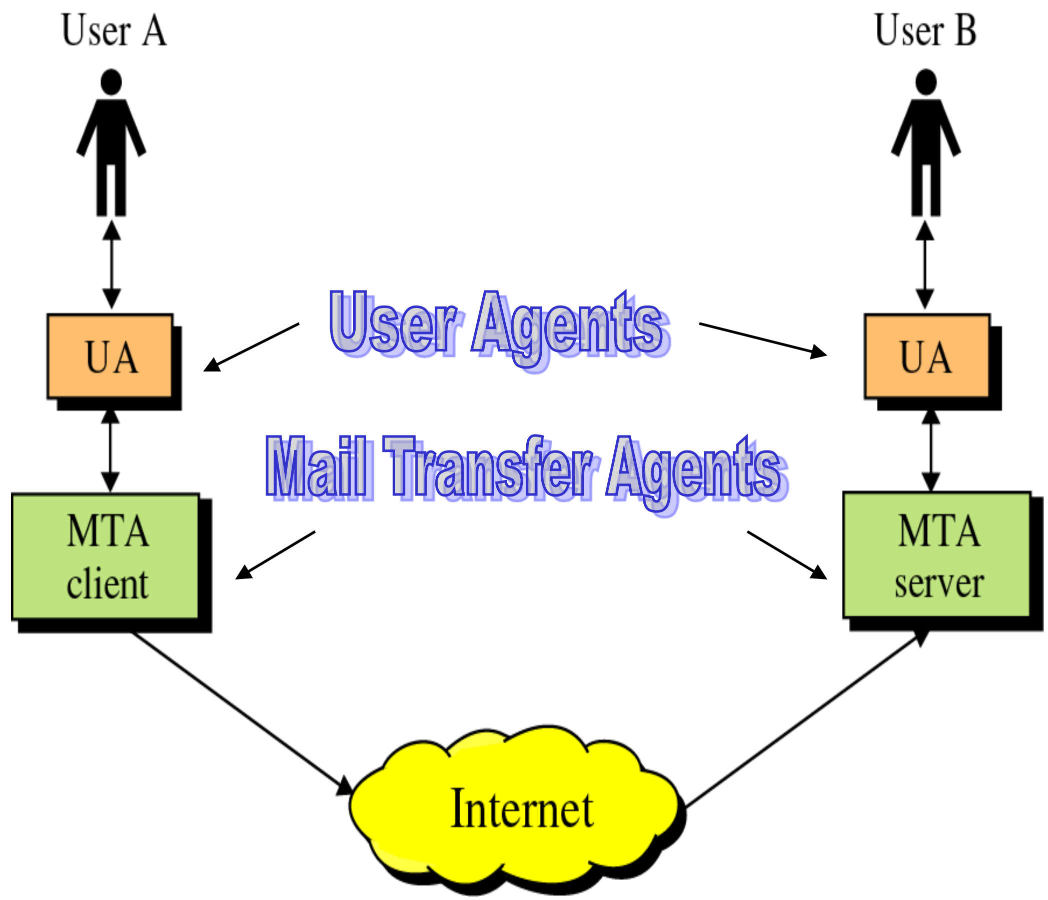
SMTP

- Protocol originated in 1982 (RFC821, Jon Postel)
- Standard message format (RFC822,2822, D. Crocker)
- Goal: To transfer mail reliably and efficiently



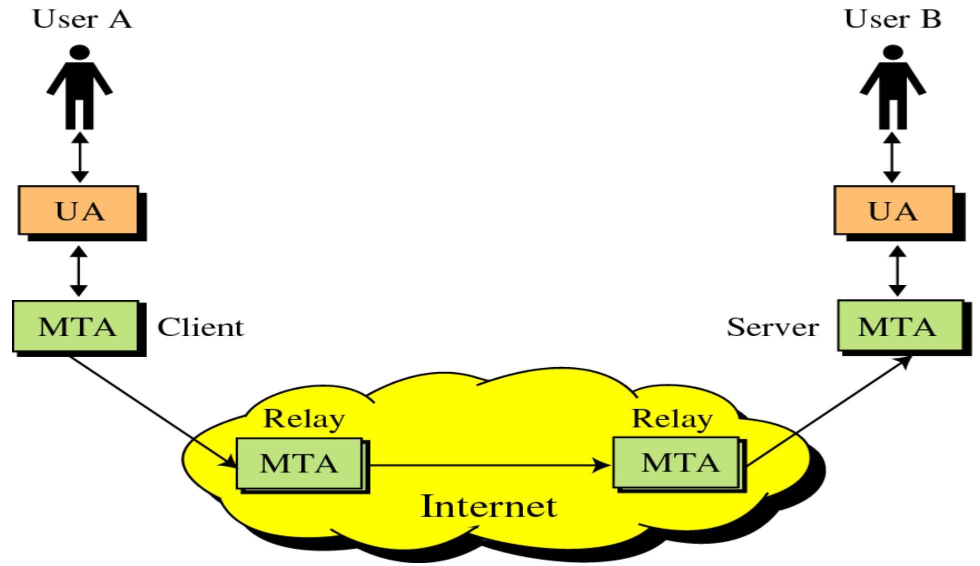
SMTP

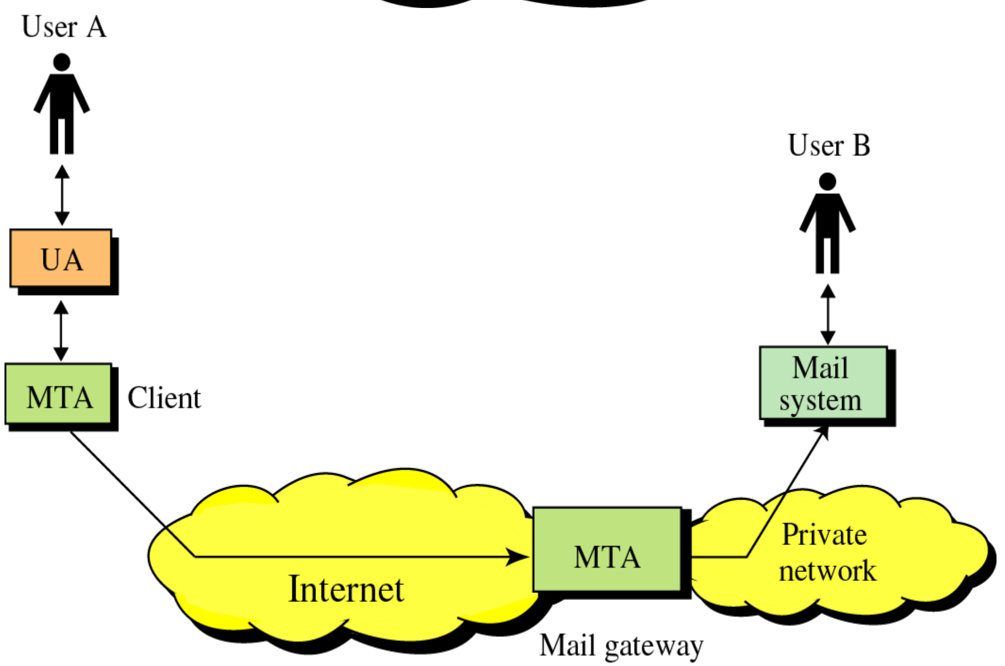
- SMTP clients and servers have two main components
  - User Agents – Prepares the message, encloses it in an envelope. (ex. Thunderbird, Eudora)
  - Mail Transfer Agent – Transfers the mail across the internet (ex. Sendmail, Exim)
  - Analogous to the postal system in many ways



SMTP

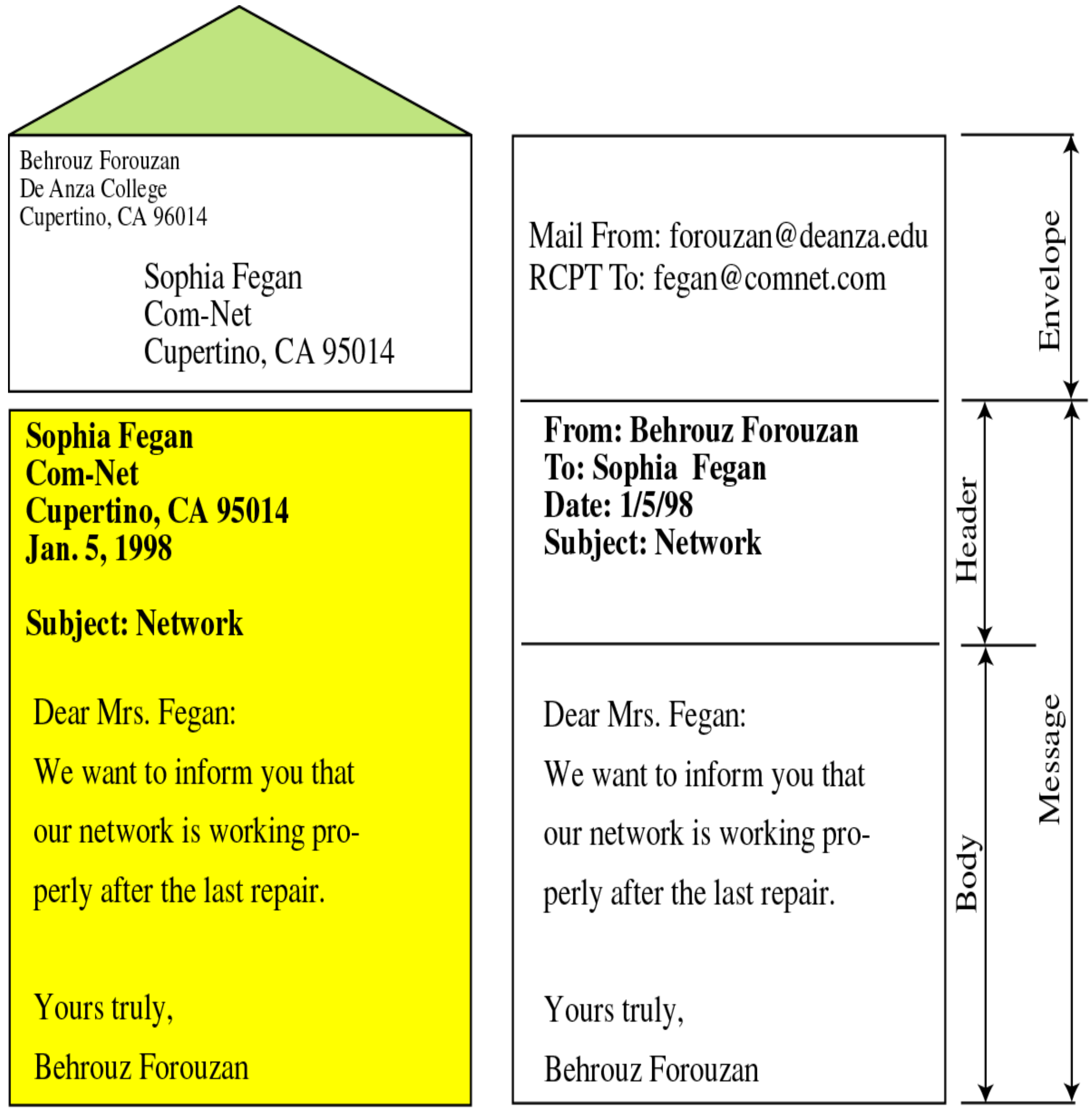
- SMTP also allows the use of other MTAs to relay the mail
- Mail Gateways are used to relay mail prepared by a protocol other than SMTP and convert it to SMTP





Format of an email

- Mail is a text file
- Envelope –
  - sender address
  - receiver address
  - other information
- Message –
  - Mail Header – defines the sender, the receiver, the subject of the message, and other information
  - Mail Body – Contains the actual information in the message



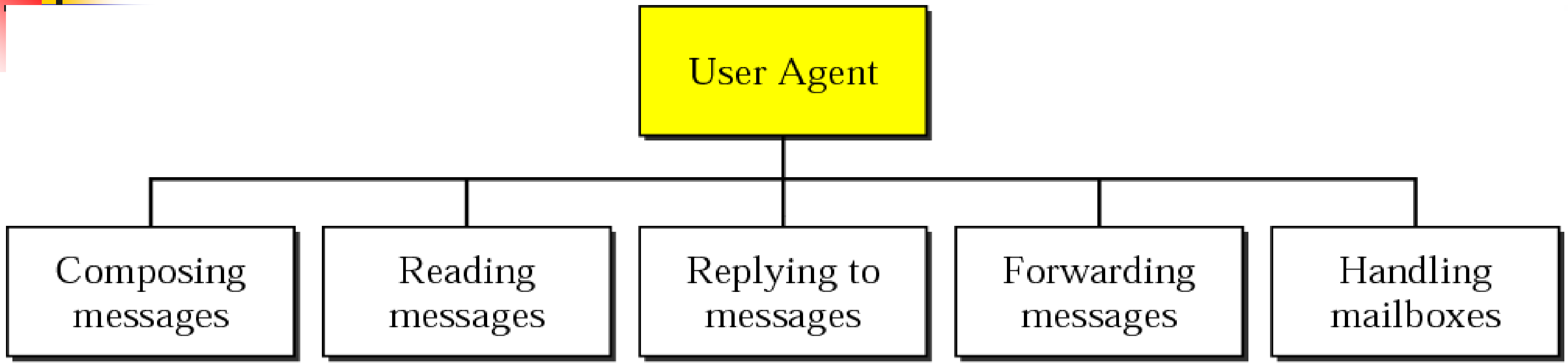
20.2 USER AGENT

The user agent (UA) First component of electronic mail.(mail reader). provides service to the user to make the process of sending and receiving a message easier.

The topics discussed in this section include:

Services Provided by a User Agent
User Agent Types
Sending Mail
Receiving Mail
Addresses
Mailing List
MIME

Figure 20.6 *User agent*



Compose :compose message to be sent, may provide template, editor
Reading: read incoming messages. check the mail(size,new,sender,subject)
Replying: to original sender or selected
Forwarding: send message to third party
Handling: create two mail box in & out handle by UA, boxes keep message deleted by user.

Two types of UA

1. Command-driven **2. GUI-Based**

Note:

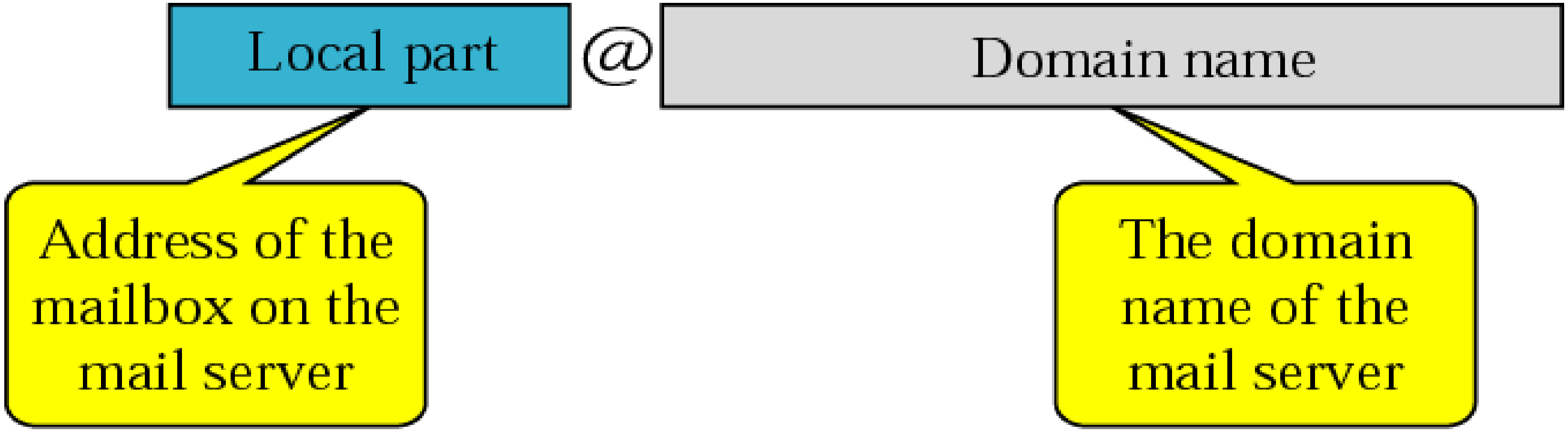
Some examples of command-driven user agents are mail, pine, and elm

Note:

Some examples of GUI-based user agents are Eudora, Outlook, and Netscape.

Figure 20.8 *Email address*

To deliver mail, mail handling system use an addressing system

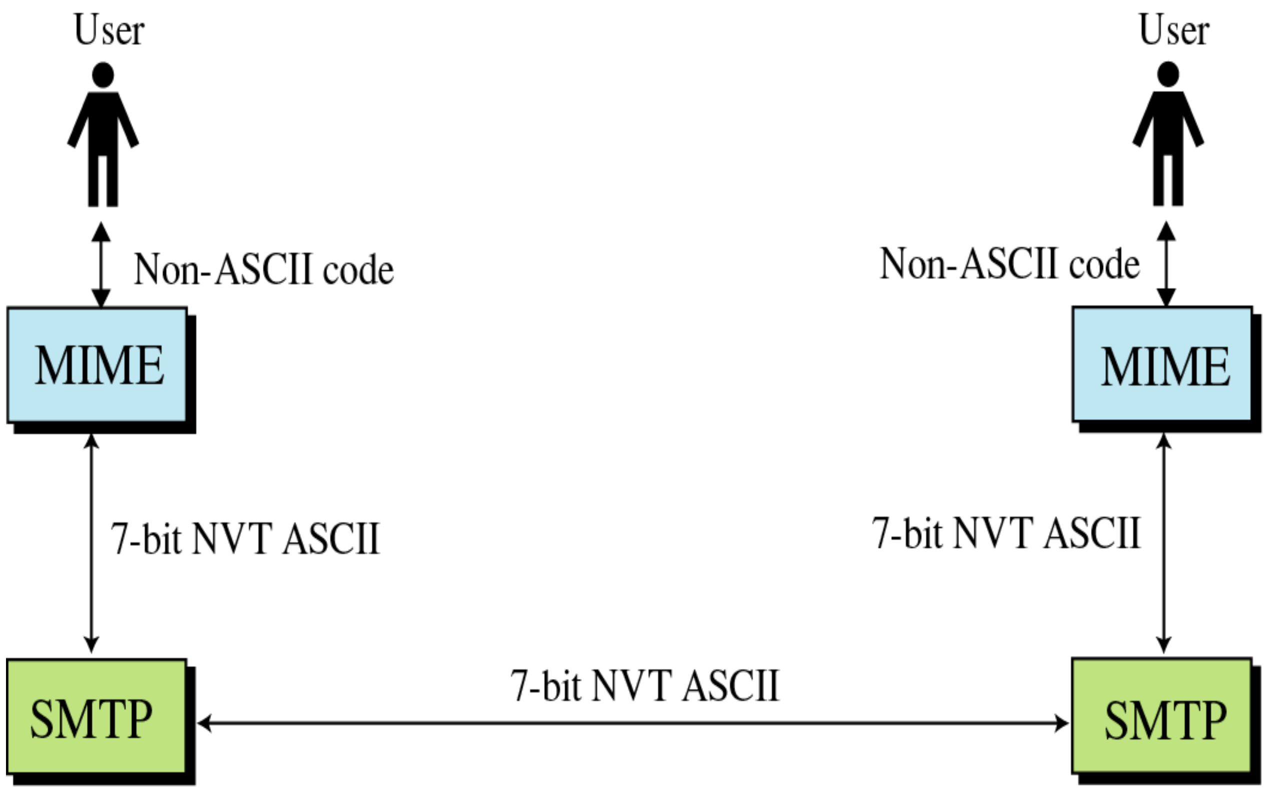


Limitations in SMTP

- Only uses 7 -bit ASCII format
  - How to represent other data types?
- No authentication mechanisms
- Messages are sent un-encrypted
- Susceptible to misuse (Spamming, faking sender address)

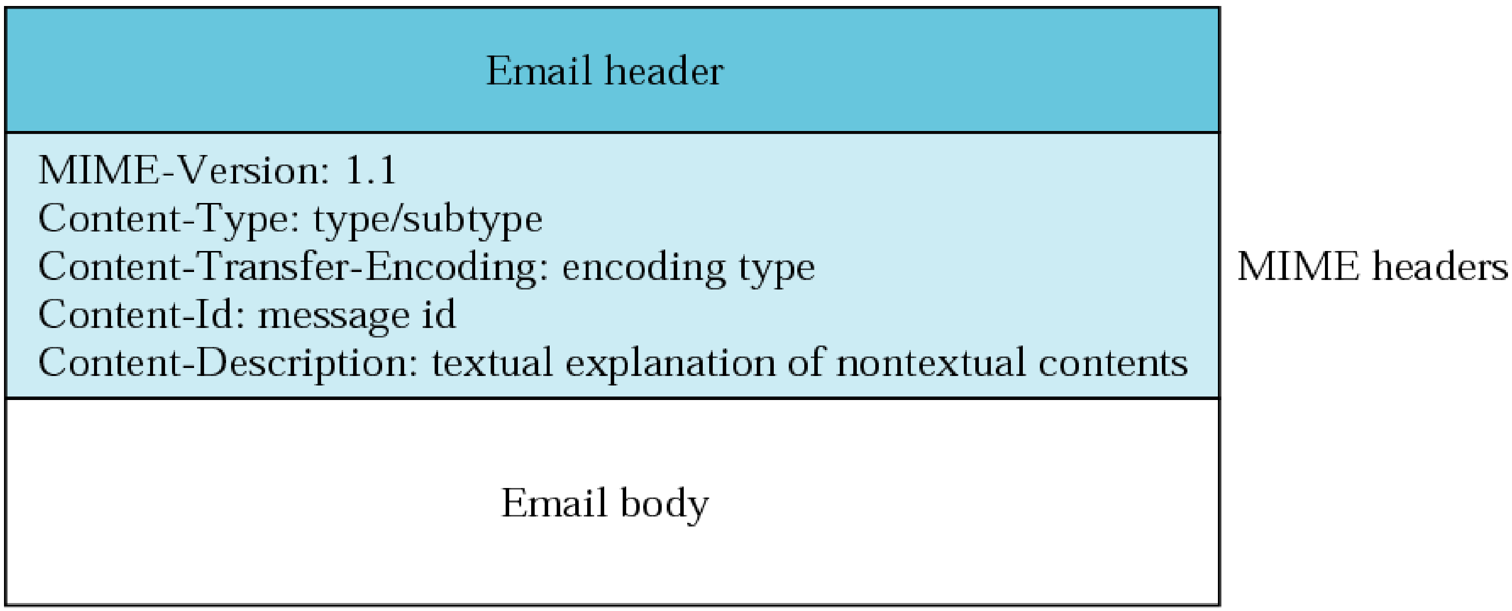
Solution: SMTP extensions

- MIME – Multipurpose Internet Mail Extensions
  - Transforms non-ASCII data to ASCII data(supplementary protocol)
    - Text
    - Application
    - Image
    - Audio
    - Video



RFC 1425, 1426, 1521

Figure 20.10 *MIME header*



MIME Headers (cont'd)

- Content-Type – Type of data used in the Body
  - Text: plain, unformatted text; HTML
  - Multipart: Body contains different data types
  - Message: Body contains a whole, part, or pointer to a message
  - Image: Message contains a static image (JPEG, GIF)
  - Video: Message contains an animated image (MPEG)
  - Audio: Message contains a basic sound sample (8kHz)
  - Application: Message is of data type not previously defined

- Content-Transfer-Encoding – How to encode the message
  - 7 bit – no encoding needed
  - 8 bit – Non-ASCII, short lines
  - Binary – Non-ASCII, unlimited length lines
  - Base64 – 6 bit blocks encoded into 8-bit ASCII
  - Quoted-printable – send non-ASCII characters as 3 ASCII characters, =##, ## is the hex representation of the byte

Figure 20.11 *Base64*

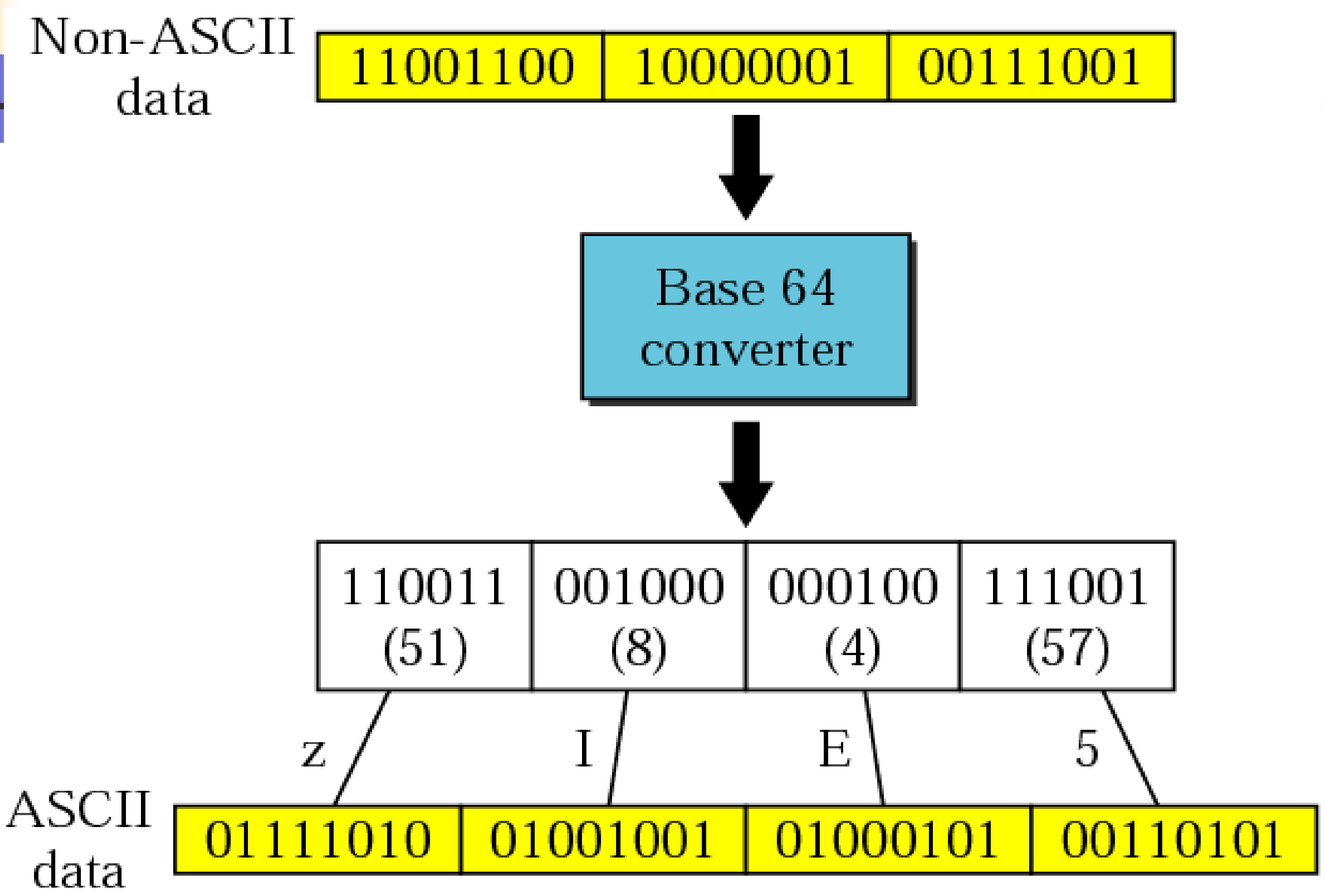
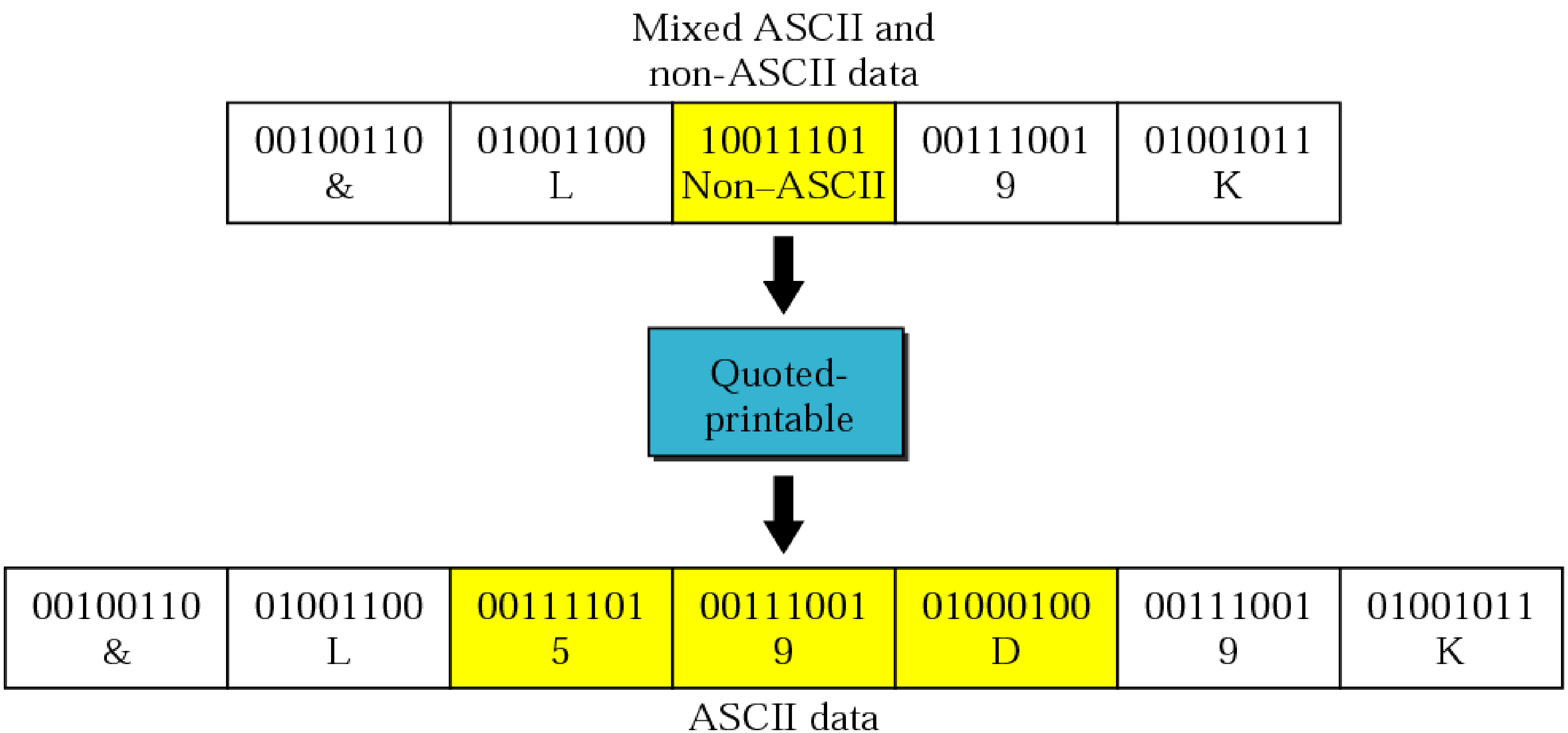


Table 20.3 Base64 encoding table

Value	*Code*	*Value*	*Code*	*Value*	*Code*	*Value*	*Code*	*Value*	*Code*	*Value*	*Code*
0	**A**	11	**L**	22	**W**	33	**h**	44	**s**	55	**3**
1	**B**	12	**M**	23	**X**	34	**i**	45	**t**	56	**4**
2	**C**	13	**N**	24	**Y**	35	**j**	46	**u**	57	**5**
3	**D**	14	**O**	25	**Z**	36	**k**	47	**v**	58	**6**
4	**E**	15	**P**	26	**a**	37	**l**	48	**w**	59	**7**
5	**F**	16	**Q**	27	**b**	38	**m**	49	**x**	60	**8**
6	**G**	17	**R**	28	**c**	39	**n**	50	**y**	61	**9**
7	**H**	18	**S**	29	**d**	40	**o**	51	**z**	62	+
8	**I**	19	**T**	30	**e**	41	**p**	52	**0**	63	/
9	**J**	20	**U**	31	**f**	42	**q**	53	**1**		
10	**K**	21	**V**	32	**g**	43	**r**	54	**2**		

Figure 20.12 *Quoted-printable*



20.3 MESSAGE TRANSFER AGENT: SMTP

The actual mail transfer requires message transfer agents (MTAs). The protocol that defines the MTA client and server in the Internet is called Simple Mail Transfer Protocol (SMTP).

The topics discussed in this section include:

Commands and Responses
Mail Transfer Phases

Figure 26.16 *SMTP range*

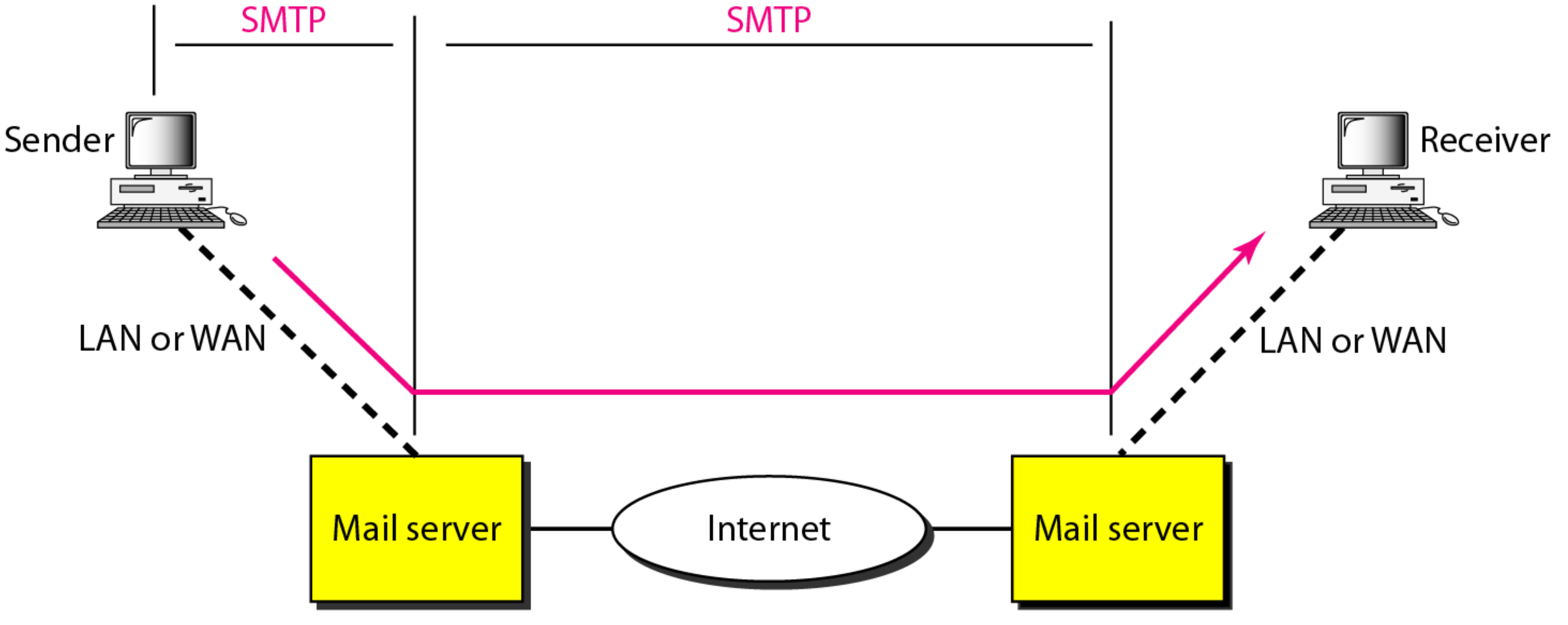


Figure 20.14 *Commands and responses*

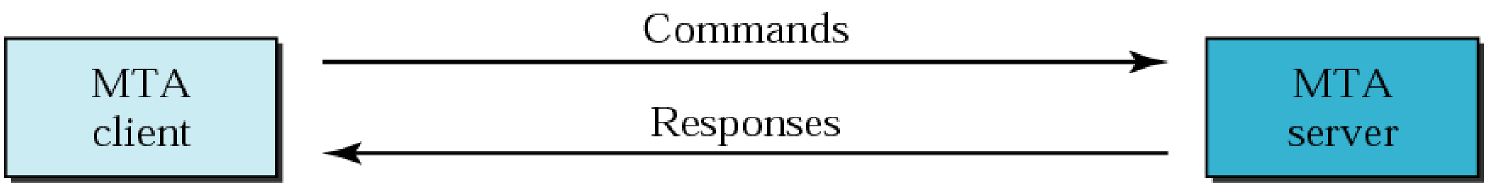


Figure 20.15 *Command format*

Keyword: argument(s)

Table 20.4 Commands

Keyword	*Argument(s)*
HELO	Sender's host name
MAIL FROM	Sender of the message
RCPT TO	Intended recipient of the message
DATA	Body of the mail
QUIT	
RSET	
VRFY	Name of recipient to be verified
NOOP	
TURN	
EXPN	Mailing list to be expanded
HELP	Command name
SEND FROM	Intended recipient of the message
SMOL FROM	Intended recipient of the message
SMAL FROM	Intended recipient of the message

Table 20.5 Responses

Code	*Description*
	Positive Completion Reply
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Request command completed
251	User not local; the message will be forwarded
	Positive Intermediate Reply
354	Start mail input
	Transient Negative Completion Reply
421	Service not available
450	Mailbox not available
451	Command aborted: local error
452	Command aborted; insufficient storage

Table 20.5 Responses (Continued)

Permanent Negative Completion Reply	
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Command is not executed; mailbox unavailable
551	User not local
552	Requested action aborted; exceeded storage location
553	Requested action not taken; mailbox name not allowed
554	Transaction failed

Figure 23.10 *Connection establishment*

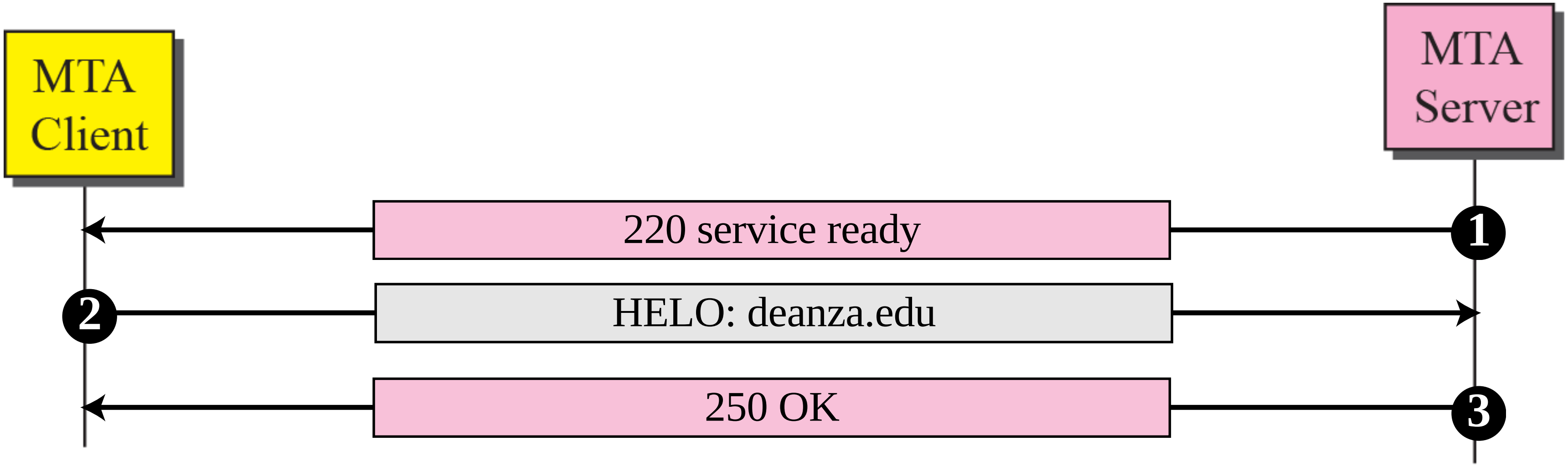


Figure 23.11 *Message transfer*

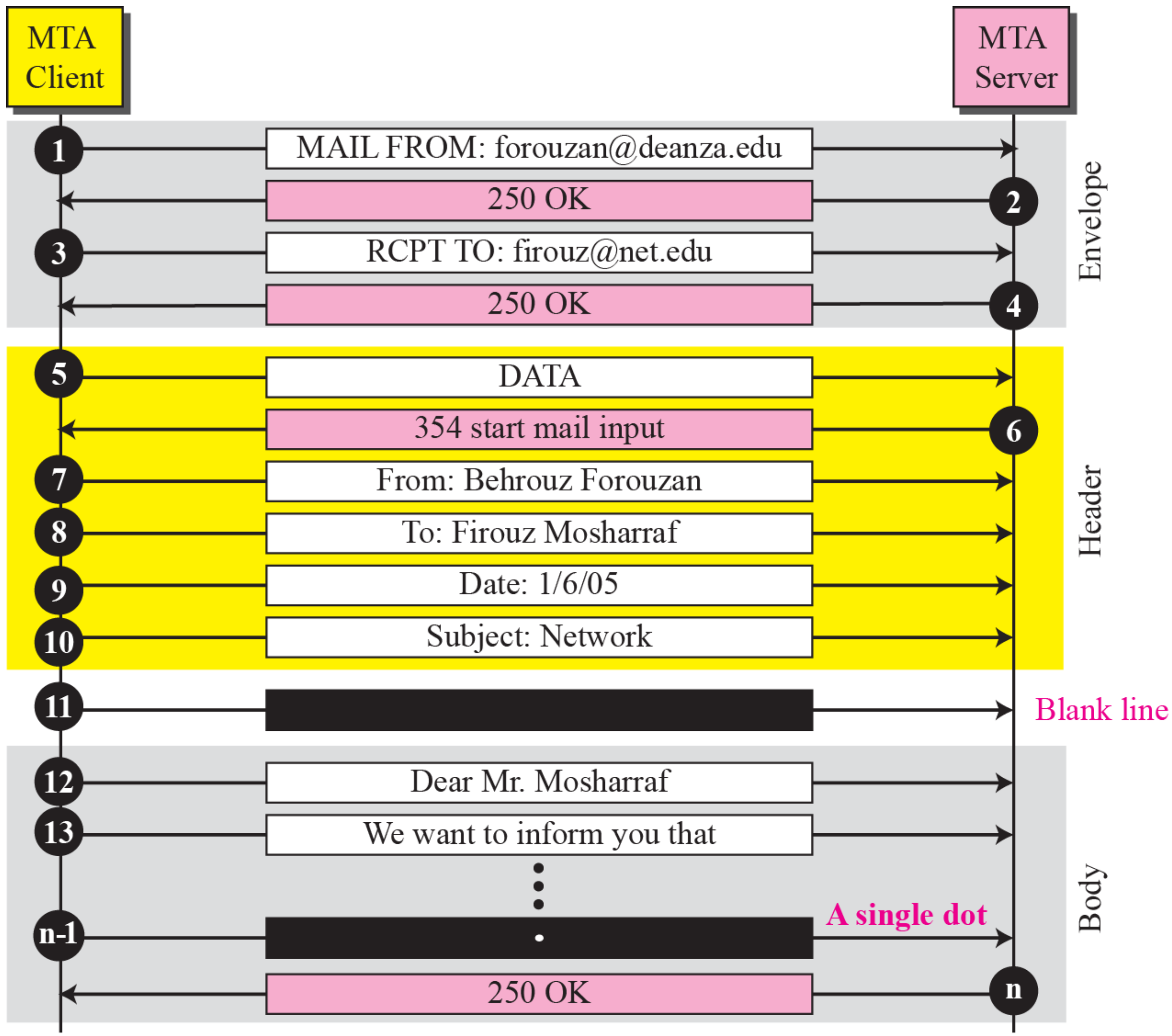


Figure 23.12 *Connection termination*

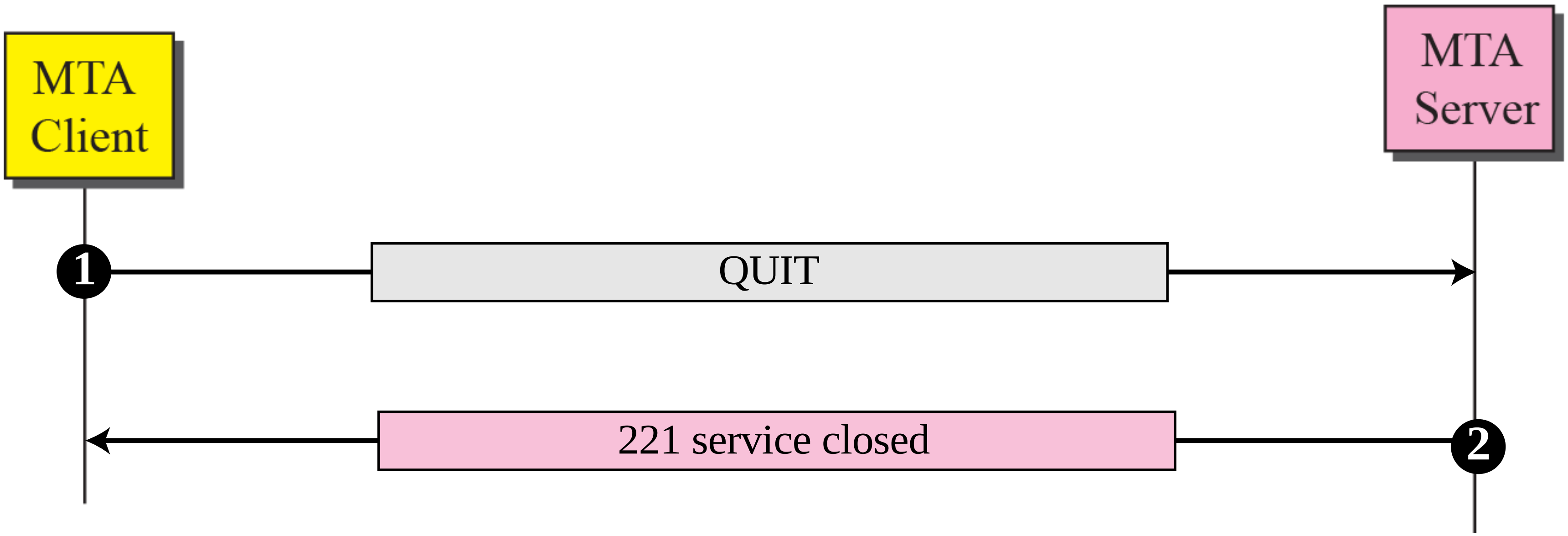
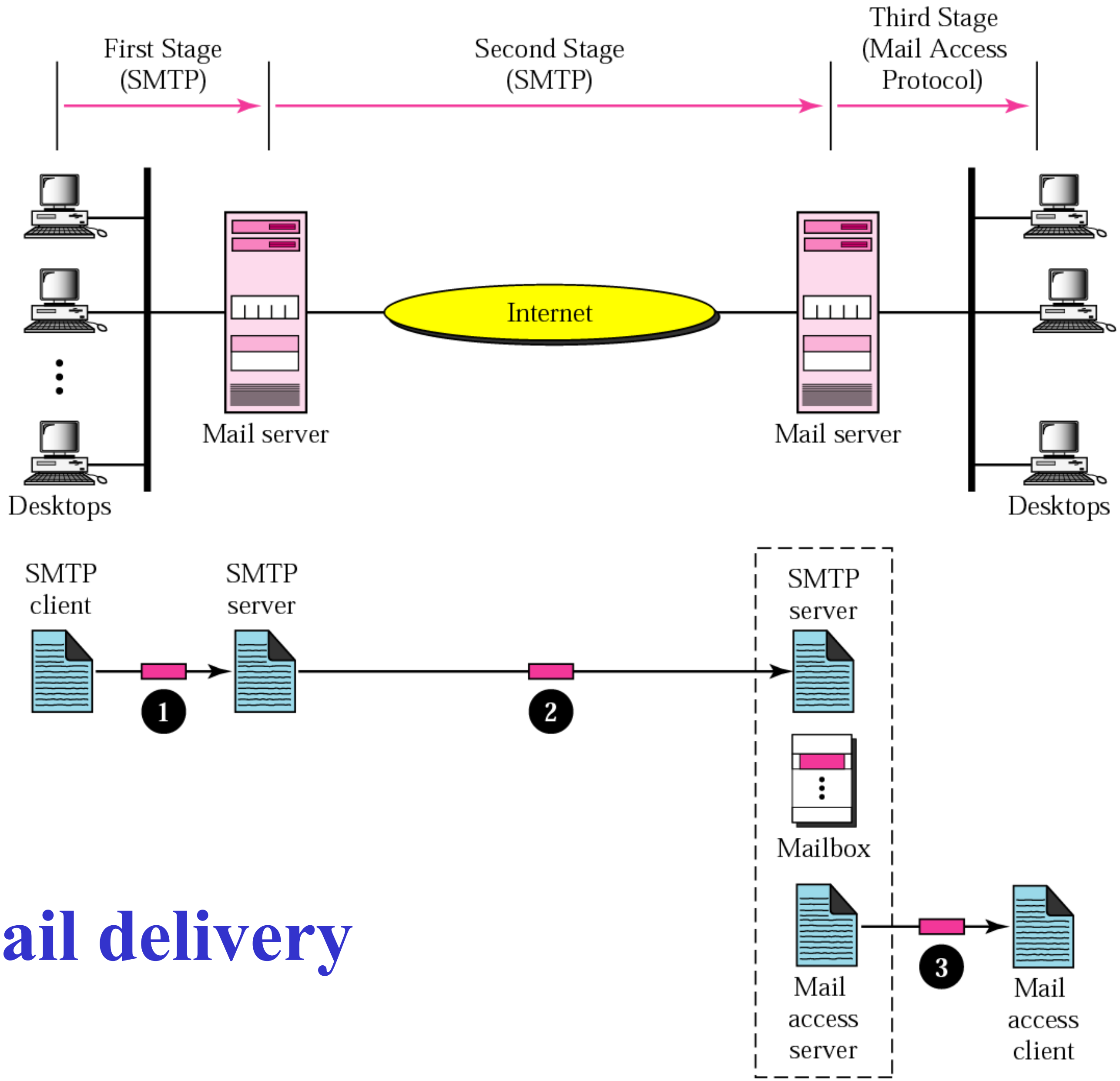


Figure 22-22



Email delivery

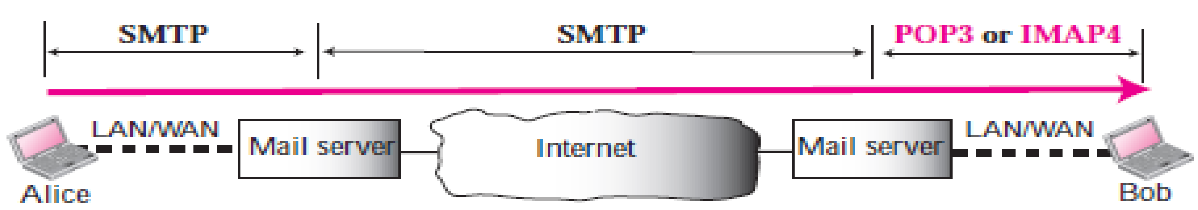
EMAIL delivery

First stage: email goes from user agent to local server….store until be sent the user agent uses SMTP client and server uses SMTP server software.

Second stage: in this stage email is relayed by local server to remote server not to remote user agent. Stored in mailbox of the user.

Third stage: the remote user agent uses a mail access protocol such as POP3 or IMAP4 .

Figure 23.13 *POP3 and IMAP4*



22.10

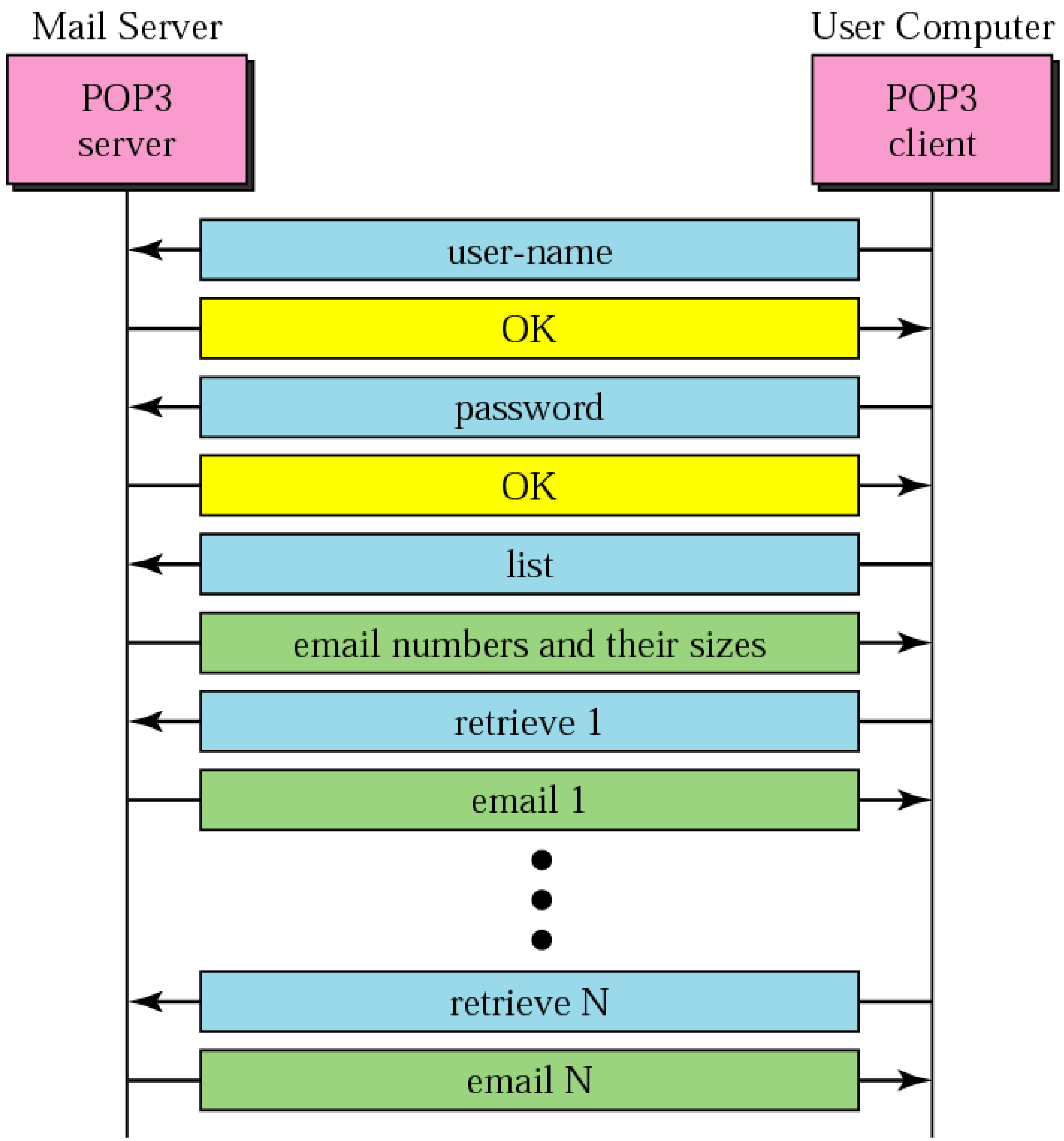
MAIL ACCESS PROTOCOLS

POP3

is simple and limited in functionality. The client POP3 software is installed on the recipient computer; the server POP3 software is installed on the mail server.
Mail access starts with the client when the user needs to download its e-mail from
the mailbox on the mail server.
The client opens a connection to the server on TCP port110.
It then sends its user name and password to access the mailbox.
The user can then list and retrieve the mail messages, one by one.

Figure 22-22

POP3(post office protocol)



IMAP4(Internet Mail Access Protocol, version 4)

IMAP4 is similar to POP3, but it has more features; more powerful and more complex. POP3 is deficient in several ways.

- ❖ It does not allow the user to organize her mail on the server;
- ❖ the user cannot have different folders on the server. (Of course, the user can create folders on her own computer.) In addition, POP3
- ❖ does not allow the user to partially check the contents of the mail before downloading.

IMAP4 provides the following extra functions:

- ❑ A user can check the e-mail header prior to downloading.
- ❑ A user can search the contents of the e-mail for a specific string of characters prior to downloading.
- ❑ A user can partially download e-mail. This is especially useful if bandwidth is limited and the e-mail contains multimedia with high bandwidth requirements.
- ❑ A user can create, delete, or rename mailboxes on the mail server.
- ❑ A user can create a hierarchy of mailboxes in a folder for e-mail storage.

FILE TRANSFER

Transferring files from one computer to another is one of the most common tasks expected from a networking or internetworking environment. As a matter of fact, the greatest volume of data exchange in the Internet today is due to file transfer.

Topics discussed in this section:

File Transfer Protocol (FTP)
Anonymous FTP

FILE TRANSFER

- File Transfer Protocol (FTP) is the standard mechanism provided by *TCP/IP for* copying a file from one host to another.
- **Problems:**

1. Different name conventions.
2. Different directory structure.
3. Different ways to represent data and text.

FILE TRANSFER

- FTP differs from other client/server applications in that it establishes two connections between the hosts.

- One connection is used for data transfer, the other for control information

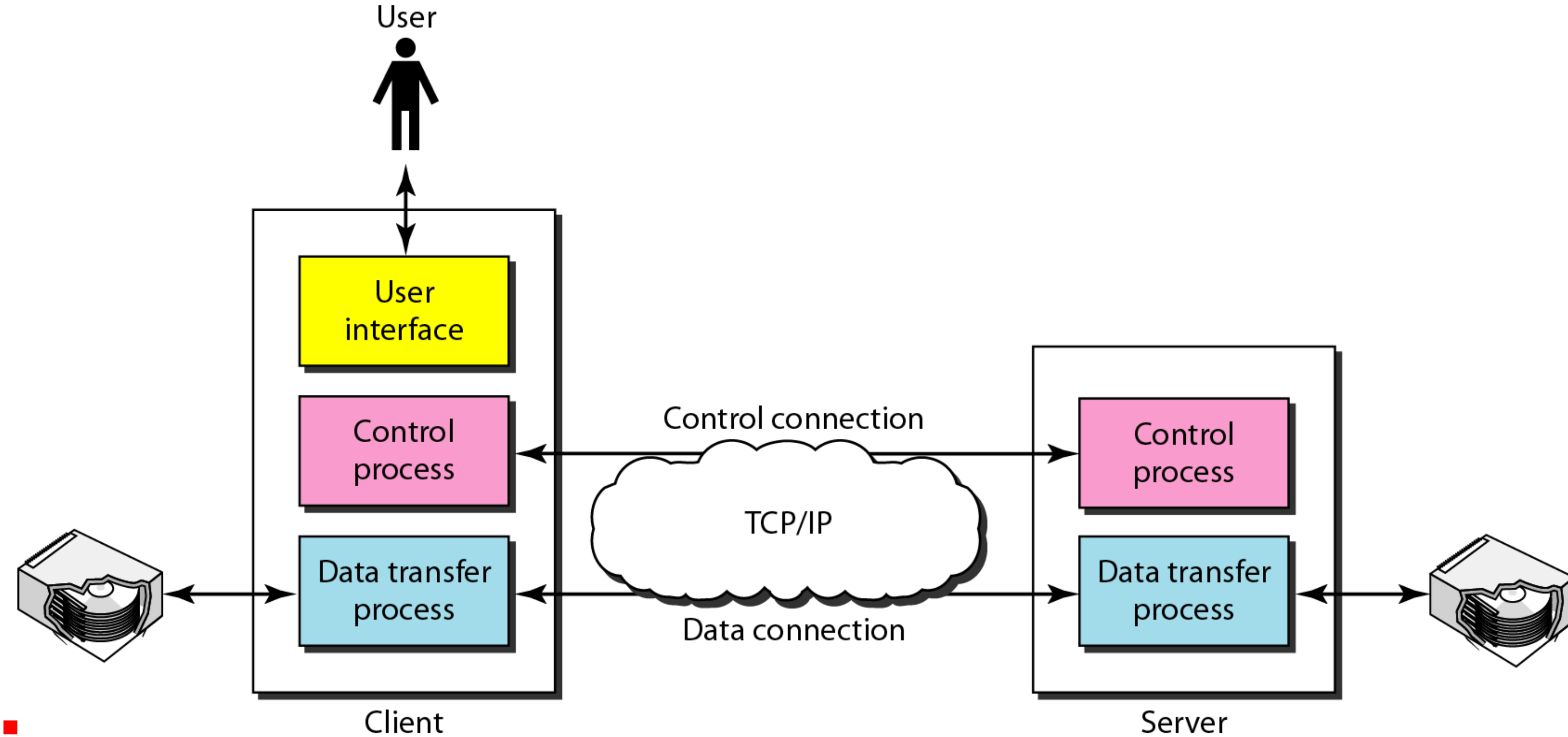
Note

FTP uses the services of TCP. It needs two TCP connections.

The well-known port 21 is used for the control connection and the well-known port 20 for the data connection.

Figure 26.21 *FTP*

The client has three components: user interface, client control process, and the client data transfer process. The server has two components: the server control process and the server data transfer process.



Note

The control connection remains connected during the entire interactive FTP session. The data connection is opened and then closed for each file transferred. It opens each time commands that involve transferring files are used, and it closes when the file is transferred.

Figure 26.22 *Using the control connection*

FTP uses the same approach as SMTP to communicate across the control connection. It uses the 7-bit ASCII character set

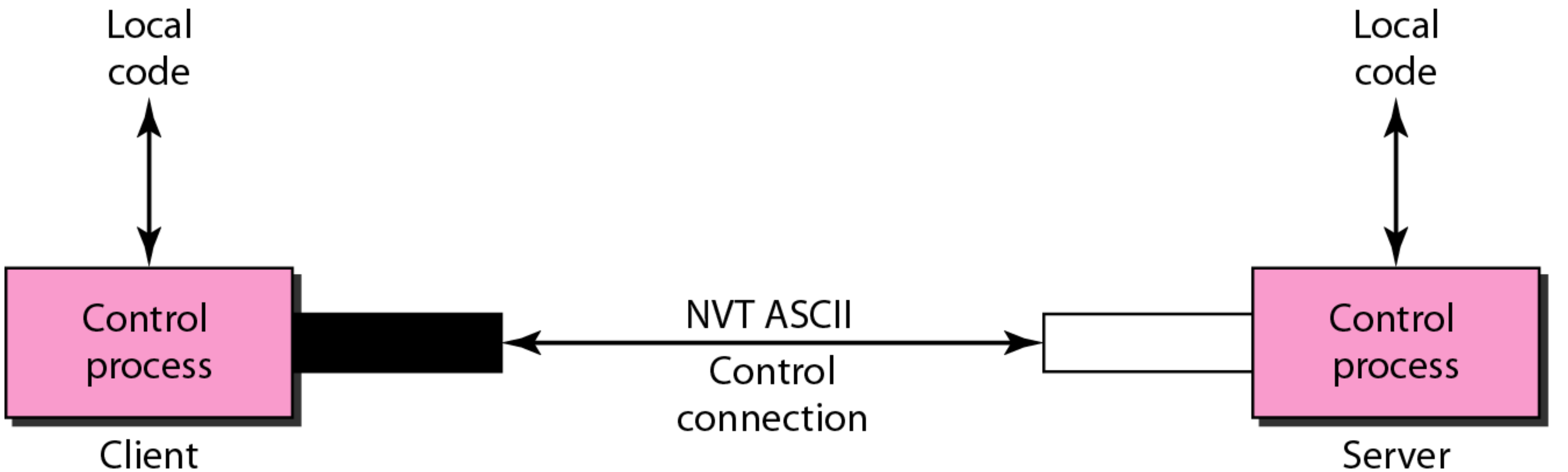
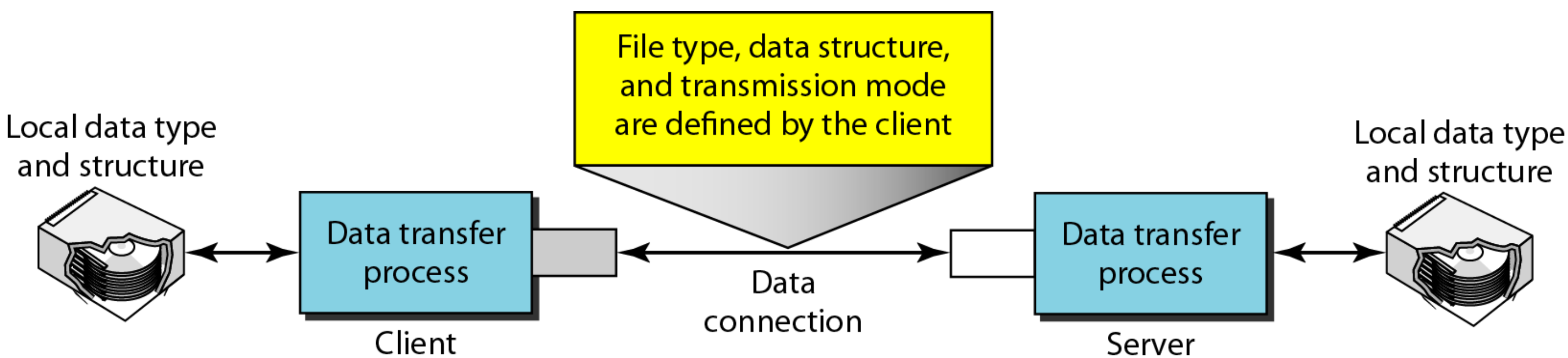


Figure 26.23 *Using the data connection*

The client must define the type of file to be transferred, the structure of the data, and the transmission mode. Before sending the file through the data connection, we prepare for transmission through the control connection. The heterogeneity problem is resolved by defining three attributes of communication: file type, data structure, and transmission mode



File Type FTP can transfer one of the following file types across the data connection:an ASCII file, EBCDIC file, or image file.

Data Structure FTP can transfer a file across the data connection by using one of the file structure, record structure, and page structure.

Transmission Mode FTP can transfer a file across the data connection by using one of the following three transmission modes: stream mode, block mode, and compressed mode

The following shows an actual FTP session for retrieving a list of items in a directory. The colored lines show the responses from the server control connection; the black lines show the commands sent by the client. The lines in white with a black background show data transfer.

1. After the control connection is created, the FTP server sends the 220 response.
2. The client sends its name.
3. The server responds with 331.

4. **The client sends the password (not shown).**
5. **The server responds with 230 (user log-in is OK).**
6. **The client sends the list command (ls reports) to find the list of files on the directory named report.**
7. **Now the server responds with 150 and opens the data connection.**
8. **The server then sends the list of the files or directories on the data connection.**
9. **The client sends a QUIT command.**
10. **The server responds with 221.**

Example 26.4 (continued)

```
$ ftp voyager.deanza.fhda.edu
Connected to voyager.deanza.fhda.edu.
220 (vsFTPd 1.2.1)
530 Please login with USER and PASS.
Name (voyager.deanza.fhda.edu:forouzan): forouzan
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls reports
227 Entering Passive Mode (153,18,17,11,238,169)
150 Here comes the directory listing.
drwxr-xr-x   2 3027    411        4096 Sep 24  2002 business
drwxr-xr-x   2 3027    411        4096 Sep 24  2002 personal
drwxr-xr-x   2 3027    411        4096 Sep 24  2002 school
226 Directory send OK.
ftp> quit
221 Goodbye.
```

Note

The control connection remains connected during the entire interactive FTP session. The data connection is opened and then closed for each file transferred. It opens each time commands that involve transferring files are used, and it closes when the file is transferred.

Note

Anonymous FTP

Some sites have a set of files available for public access, to enable anonymous FTP. To access these files, a user does not need to have an account or password. Instead, the user can use *anonymous as the user name and guest as the password.*

Questions ?